3-mavzu. Teskari matritsa va matritsaning rangi.

Reja

3.1. Teskari matritsa.

3.2. Matritsaning rangi

3.3. Matritsani iqtisodda qo'llanilishi.

Tayanch iboralar: *minor, algebrail to'ldiruvchi, elementar almashtirishlar*

3.1. Teskari matritsa haqida teorema.

Teskari matritsa formulasini topishdan oldin ba'zi zarur tushunchalar bilan tanishamiz.

Transponirlangan matritsa. Marisa mos satrlarini uning mos ustunlariga almashtirishdan hosil bo'lgan matritsaga transponoirlangan matritsa deyiladi. A matritsaning trasponirlangani A^T kabi belgilanadi.

Masalan, $A=\begin{bmatrix} 5 & 20 \\ 16 & 9 \\ 12 & 6 \end{bmatrix}$ uchun transponirlangan matritsa $A^T=\begin{bmatrix} 5 & 16 & 12 \\ 20 & 9 & 6 \end{bmatrix}$

Kofaktor matritsa. Matritsa har bir elementini uning algaebraik to'ldiruvchisi bilan almashtirishdan hosil bo'lgan matritsa kofaktor matritsa deyiladi va u C bilan belgilanadi.

Masalan, $A=\begin{bmatrix} 2 & 4 & 3 \\ 3 & 5 & 0 \\ 4 & 2 & 5 \end{bmatrix}$ uchun $C=\begin{bmatrix} 25 & -15 & -12 \\ -14 & -2 & 12 \\ -15 & 9 & -2 \end{bmatrix}$

Bu matritsa elementlari qanday hisoblanganini o'rganaylik. **A** matritsa $|C_{ij}|$ algebraik to'ldiruvchisini hisoblash uchun shu element joylashgan I satr va j ustu o'chirilib, hosil bo'lgan determinant $(-1)^{i+j}$ ishora aniqligida olinadi.

C matritsa elementlari quyidagicha hisoblangan

$$c_{11}=|C_{11}|=(-1)^{(1+1)}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}=(-1)^2\begin{vmatrix} 5 & 0 \\ 2 & 5 \end{vmatrix}=(25-0)=25$$

$$c_{21}=|C_{21}|=(-1)^{(2+1)}\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}=(-1)^3\begin{vmatrix} 4 & 3 \\ 2 & 5 \end{vmatrix}=(-1)(20-6)=-14$$

va h.k.

C matritsa qolgan elementlarni o'zingiz tekshiring.

Biriktirilgan matritsa. Kofaktor matritsaning trasponirlangan matritsasiga biriktirilgan matritsa deyiladi va *A* matritsaning biriktirilgan matritsasi *Adj A* kabi belgilanadi.

Masalan, $A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ matritsa uchun $AdjA=\begin{bmatrix} |C_{11}| & |C_{21}| & |C_{31}| \\ |C_{12}| & |C_{22}| & |C_{32}| \\ |C_{13}| & |C_{23}| & |C_{33}| \end{bmatrix}$

Yuqoridagi misoldagi $A=\begin{bmatrix} 2 & 4 & 3 \\ 3 & 5 & 0 \\ 4 & 2 & 5 \end{bmatrix}$ matritsa uchun kofaktor matritsa $C=\begin{bmatrix} 25 & -15 & -12 \\ -14 & -2 & 12 \\ -15 & 9 & -2 \end{bmatrix}$ va biriktirilgan matritsa $AdjA=C^T=\begin{bmatrix} 25 & -14 & -15 \\ -15 & -2 & 9 \\ -12 & 12 & -2 \end{bmatrix}$.

Teskari matritsa. Xos bo'lmagan, ya'ni $|A|$ determinant nol bo'lmagan, A matritsaning teskari matritsasi

$$A^{-1}=\frac{AdjA}{|A|}$$

formula yordamida hisoblanadi..

Misol. $A=\begin{bmatrix} 2 & 4 & 3 \\ 3 & 5 & 0 \\ 4 & 2 & 5 \end{bmatrix}$matritsaga A^{-1} teskari matritsa topilsin.

Yechish. Bu matritsa uchun yuqoridagi misolda biriktirilgan matritsa topgan edik. Bu matritsa determinantini hisoblaymiz

$$\begin{aligned} |A| &= 3\begin{vmatrix} 3 & 5 \\ 4 & 2 \end{vmatrix} - 0 + 5\begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} = \\ &= 3(6-20)+5(10-12)= \\ &= 3(-14)+5(-2)= \\ &= -42-10=-52 \end{aligned}$$

Shuning uchun,

$$\boldsymbol{AdjA}=\begin{bmatrix} 25 & -14 & -15 \\ -15 & -2 & 9 \\ -12 & 12 & -2 \end{bmatrix}$$

Bo'lgani uchun teskari matritsa quyidagi ko'rinishda bo'ladi

$$A^{-1}=\frac{AdjA}{|A|}=\frac{\begin{bmatrix} 25 & -14 & -15 \\ -15 & -2 & 9 \\ -12 & 12 & -2 \end{bmatrix}}{-52}=\begin{bmatrix} -0.48 & 0.27 & 0.29 \\ 0.29 & 0.04 & -0.17 \\ 0.27 & -0.23 & 0.04 \end{bmatrix}$$

Bu matritsani hisoblash ko'p vaqt talab etadi, lekin siz bu usul asosiyligini bilishingiz kerak, elektron jadvalda hisoblashdan oldin. Endi misol yordamida har bir etapni aniqlaylik. Soddalik uchun 2x2 o'lchovli holni qaraylik.

Misol. $A=\begin{bmatrix} 20 & 5 \\ 6 & 2 \end{bmatrix}$ matritsaga A^{-1} teskari matritsa toping.

Yechish. A matritsa to'rt elementli bo'lgani uchun, kofaktor matritsa bosh diagonal elementlari qarama-qarshi ishorali bo'ladi.

Bu matrrisaga kofaktor matritsa

$$C=\begin{bmatrix} 2 & -5 \\ -5 & 20 \end{bmatrix}$$

matritsa bo'ladi.

Bu matritsaga biriktirilgan matritsa

$$\boldsymbol{AdjA}=\begin{bmatrix} 2 & -5 \\ -5 & 20 \end{bmatrix}$$

bo'ladi.

Bu matritsa determinanti

$$|A|=20\times 2-5\times 6=40-30=10.$$

Demak bu matritsaga teskari matritsa

$$A^{-1}=\frac{AdjA}{|A|}=\frac{\begin{bmatrix} 2 & -5 \\ -6 & 20 \end{bmatrix}}{10}=\begin{bmatrix} 0.2 & -0.5 \\ -0.6 & 2 \end{bmatrix}.$$

Ta'rif. *A kvadrat matritsaga teskari matritsa deb, shunday A^{-1} matritsaga aytiladiki, uning uchun quyidagi $A\cdot A^{-1}=A^{-1}\cdot A=E$ tenglik o'rinli bo'lsin.*

Ta'rif. *Agar A matritsa uchun $|A|\neq 0$ bo'lsa, bunday matritsa xos bo'lmagan matritsa, aks holda, ya'ni $|A|=0$ bo'lsa xos matritsa deyiladi.*

Teorema. *A kvadratik matritsaga teskari A^{-1} matritsa mavjud va yagona bo'lishi uchun, uning xos bo'lmagan matritsa bo'lishi zarur va yetarlidir.*

A-matritsa xos bo'lmagan matritsa bo'lsin, ya'ni $|A|\neq 0$ bo'lsin. U holda

$$A^{-1}=\frac{1}{|A|}\begin{pmatrix} A_{11} & A_{21}\ldots A_{j1}\ldots A_{n1} \\ A_{12} & A_{22}\ldots A_{j2}\ldots A_{n2} \\ A_{1k} & A_{2k}\ldots A_{jk}\ldots A_{nk} \\ A_{1n} & A_{2n} \quad A_{jn}\ldots A_{nn} \end{pmatrix}.$$

Misol. $A=\begin{pmatrix} 1 & 2 & 1 \\ 3 & -5 & 3 \\ 2 & 7 & -1 \end{pmatrix}$ matritsaga teskari matritsani toping.

Yechish. $|A|=\begin{vmatrix} 1 & 2 & 1 \\ 3 & -5 & 3 \\ 2 & 7 & -1 \end{vmatrix}=5+12+21+10-21+6=33\cdot$ $\quad A_{11}=-16,\ A_{12}=9,\ A_{13}=31$

$A_{21}=9,\ A_{22}=-3,\ A_{23}=-3$

$A_{31}=11,\ A_{32}=0,\ A_{33}=-11$

$$A^{-1}=\frac{1}{|A|}\begin{pmatrix}-16 & 9 & 11\\ 9 & -3 & 0\\ 31 & -3 & -11\end{pmatrix}=\frac{1}{33}\begin{pmatrix}-16 & 9 & 11\\ 9 & -3 & 0\\ 31 & -3 & -11\end{pmatrix}=\begin{pmatrix}-\frac{16}{33} & \frac{3}{11} & \frac{1}{3}\\ \frac{3}{11} & -\frac{1}{11} & 0\\ \frac{31}{33} & -\frac{1}{11} & -\frac{1}{3}\end{pmatrix}$$

3.2. Matritsaning rangi

Ta'rif. *A matritsaning rangi deb uning noldan farqli minorlarining eng yuqori tartibiga aytilib, $r(A)$ orqali belgilanadi.*

Ta'rif. *Matritsa ustidagi elementar almashtirishlar deb quyidagi almashtirishlarga aytiladi:*

1. Barcha elementlari noldan iborat satrni (ustunni) tashlab yuborish.

2. Satrning (ustunning) barcha elementlarini noldan farqli songa ko'paytirish.

3. Satr (ustun) o'rinlarini almashtirish.

4. Berilgan satr (ustun) elementlariga boshqa satr (ustun) elementlarini biron songa ko'paytirib qo'shish.

5. Matritsani transponirlash.

Teorema. *Matritsa rangi uning ustida elementar almashtirishlarni bajarish natijasida o'zgarmaydi.*

Matritsa rangi uchun quyidagi xossalar o'rinli

1. $r(A+B)\le r(A)+r(B)$ 2. $r(A+B)\ge |r(A)-r(B)|$

3. $r(AB)\le \min\{r(A),r(B)\}$ 4. $r(A'A)=r(A)$

Ta'rif. *Agar* $I_1=(a_{11},a_{12},\ldots,a_{1n}),\ I_2=(a_{21},\ a_{22},\ \ldots,a_{2n}),\cdots,I_m=(a_{m1},a_{m2},\cdots,a_{mn})$ *satrlar va* $\lambda_1,\lambda_2,\ldots\lambda_m$ *sonlar uchun* $\lambda_1 I_1+\lambda_2 I_2+\cdots+\lambda_m I_m=0$ *tenglik faqat* $\lambda_1=\lambda_2=\ldots=\lambda_m=0$ *shartdagina o'rinli bo'lsa, u holda* $I_1, I_2,\ldots,I_m$ *satrlar chiziqli erkli satrlar , aks holda esa ular chiziqli bog'liq satrlar deyiladi.*

Teorema. *Matritsa uchun uning chiziqli erkli satrlarning maksimal soni chiziqli erkli ustunlarning maksimal soniga teng bo'ladi.*

Teorema. *Matritsa rangi undagi chiziqli erkli satrlarning (ustunlarning) maksimal soniga teng bo'ladi.*

Matritsa rangini "rang ta'rifi" bo'yicha hisoblash doim ham oson ish bo'lavermaydi. Matritsa rangini topish uchun matritsa ustida elementar almashtirishlar bajarish natijasida bu matritsani uchburchak yoki trapetsiya ko'rinishiga olib kelish mumkin.

Misol. Berilgan matritsalarning rangini toping. $A=\begin{pmatrix}5 & 2 & 1 & 3\\ 3 & -1 & 1 & 2\\ 1 & 2 & 4 & 8\\ 3 & -1 & 1 & 2\end{pmatrix}$

Yechish.

-1	-4	-1

$$\begin{pmatrix} 5 & 2 & 1 & 3 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 4 & 8 \\ 3 & -1 & 1 & 2 \end{pmatrix} \begin{matrix} \\ \\ + \quad + \\ \\ \end{matrix} \rightarrow \begin{pmatrix} 5 & 2 & 1 & 3 \\ -2 & -3 & 0 & -1 \\ -19 & -6 & 0 & -4 \\ -2 & -3 & 0 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 5 & 3 \\ 0 & -3 & -2 & -1 \\ 0 & -6 & -19 & -4 \\ 0 & -3 & -2 & -1 \end{pmatrix} \begin{matrix} \boxed{-2} \boxed{-1} \\ \\ + \quad + \\ \\ \end{matrix}$$

$$\rightarrow \begin{pmatrix} 1 & 2 & 5 & 3 \\ 0 & -3 & -2 & -1 \\ 0 & 0 & -15 & -2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Demak, *rang*(*A*) = 3.

3.3. Matritsani iqtisodda qo'llanilishi.

Misol 1. Neftga bo'lan talab (million barrelda) ni **q** = **βx** modelida tushuntirish mumkin va bunda

$$\beta = \begin{bmatrix} \beta_0 & \beta_1 & \beta_2 & \beta_3 & \beta_4 & \beta_5 \end{bmatrix} = \begin{bmatrix} 4.2 & -0.1 & 0.4 & 0.2 & -0.1 & 0.2 \end{bmatrix}$$

bo'lsin,deb faraz qilaylik.

Tavsiflovchi o'zgaruvchilar vektori

$$x = \begin{bmatrix} 1 \\ x_1^t \\ x_2^t \\ x_3^t \\ x_4^t \\ x_5^t \end{bmatrix} = \begin{bmatrix} \text{Constant} \\ \text{Price} \\ \text{Income} \\ \text{Price of substitute} \\ \text{Price of complement} \\ \text{Population (in m.)} \end{bmatrix} = \begin{bmatrix} 1 \\ 30 \\ 18.5 \\ 52 \\ 12.8 \\ 61 \end{bmatrix}$$

bo'lganda neftga bo'lgan talabni hisoblang.

Yechish. Neftga bo'lgan talabni quyidagicha hisoblanadi

$$q = \beta x = \begin{bmatrix} 4.2 & -0.1 & 0.4 & 0.2 & -0.1 & 0.2 \end{bmatrix} \begin{bmatrix} 1 \\ 30 \\ 18.5 \\ 52 \\ 12.8 \\ 61 \end{bmatrix} = \begin{bmatrix} 29.92 \end{bmatrix}$$

Shunday qilib javob 29,92 million barrel.

Misol 2. Telefon apparatlarini ta'mirlovchi usta 70% telefonlarni past darajada, 20% o'rta darajada va 10% to'liq ta'mirdan chiqardi. Statistik ma'lumotlarga ko'ra 70% past darajada ta'mirlangan telefonlarni bir yildan keyin qayta 10% past darajada, 60% o'rta darajada, 30% ni to'liq ta'mirlashadi. O'rta darajada ta'mirlangan telefonlarni bir yildan keyin qayta 20% past darajada, 50% o'rta, 30% ni to'liq ta'mirlashadi. To'liq ta'mirlangan telefonlarni bir yildan keyin qayta 60% past darajada, 40% o'rta darajada ta'mirlashadi. Agar masala sharti shu

tarzda davom etsa 1, 2, 3 – yillardan keyingi har bir darajada ta'mirlangan telefonlar ulushini aniqlashda matritsalar algebrasidan foydalanish qulay.

$$X_0 = (0{,}7 \quad 0{,}2 \quad 0{,}1)$$

$$A = \begin{pmatrix} 0{,}1 & 0{,}6 & 0{,}3 \\ 0{,}2 & 0{,}5 & 0{,}3 \\ 0{,}6 & 0{,}4 & 0 \end{pmatrix}, \qquad \begin{aligned} X_1 &= X_0 \cdot A = (0{,}17 \quad 0{,}56 \quad 0{,}27) \\ X_2 &= X_1 \cdot A = (0{,}291 \quad 0{,}490 \quad 0{,}219) \\ X_3 &= X_2 \cdot A = (0{,}2585 \quad 0{,}5072 \quad 0{,}2343) \end{aligned}$$

Mavzuni mustaxkamlash test savollari.

1. Qachon kvadrat matritsaning teskari matritsasi mavjud bo'ladi?
A) Barcha elementlari musbat bo'lsa
B) Determinanti 0 ga teng bo'lmasa
C) Barcha elementlari teng bo'lsa
D) Deteminanti manfiy bo'lsa
To'g'ri javob: B

2. Teskari matritsa A^{-1}qanday xossalarga ega?
A) $A \cdot A^{-1} = A$
B) $A^{-1} \cdot A = 0$
C) $A \cdot A^{-1} = I$
D) $A \cdot A^{-1} = A^2$
To'g'ri javob: C

3. Teskari matritsani topishda qaysi usul ishlatiladi?
A) Gauss usuli
B) Determinantni nollash
C) Minorlar bo'yicha yoyish
D) Diagonalga bo'lish
To'g'ri javob: A

4. Quyidagilardan qaysi biri teskari matritsa topishning sharti hisoblanadi?
A) Matritsa nolli bo'lishi kerak
B) Matritsa simmetrik bo'lishi kerak
C) Matritsaning determinanti 0 bo'lmasligi kerak
D) Matritsa to'rtburchak shaklda bo'lishi kerak
To'g'ri javob: C

5. Matritsaning rangi deganda nima tushuniladi?
A) Diagonal elementlari soni
B) Matritsadagi barcha elementlarning yig'indisi
C) Nolli bo'lmagan maksimal o'lchamli minorning tartibi
D) Qator va ustunlar sonining ko'paytmasi
To'g'ri javob: C

6. Rangni aniqlashda qanday usuldan foydalaniladi?
A) Transpozitsiya usuli
B) Gauss metodidan
C) Diagonal metodidan
D) Determinantlarni yig‘ish
To‘g‘ri javob: B

7. Agar matritsaning rangi 3 bo‘lsa, bu nimani anglatadi?
A) Matritsa 3 elementdan iborat
B) 3 ta qator mavjud
C) 3×3 o‘lchamli nolli bo‘lmagan minor mavjud
D) Barcha elementlari 3 ga teng
To‘g‘ri javob: C

8. Matritsaning rangi maksimal darajada bo‘lishi uchun qanday shart kerak?
A) Faqat bitta ustun bo‘lishi kerak
B) Barcha minorlar nollarga teng bo‘lishi kerak
C) Matritsaning barcha qatorlari chiziqli bog‘liq bo‘lishi kerak
D) Matritsaning qatorlari chiziqli mustaqil bo‘lishi kerak
To‘g‘ri javob: D

9. Matritsalar iqtisodiyotda qanday masalalarda qo‘llaniladi?
A) Tashqi savdo balansini tahlil qilishda
B) Iqlim o‘zgarishini kuzatishda
C) Yadro fizikasi modellarida
D) Geografik koordinatalarni hisoblashda
To‘g‘ri javob: A

10. Teskari matritsa iqtisodiyotda qanday muammoni hal qilishda ishlatiladi?
A) Daromad va xarajatlar tahlilida
B) Ehtimollar nazariyasida
C) Ma’lumotlarni kodlashda
D) Matnli axborotni tarjima qilishda
To‘g‘ri javob: A